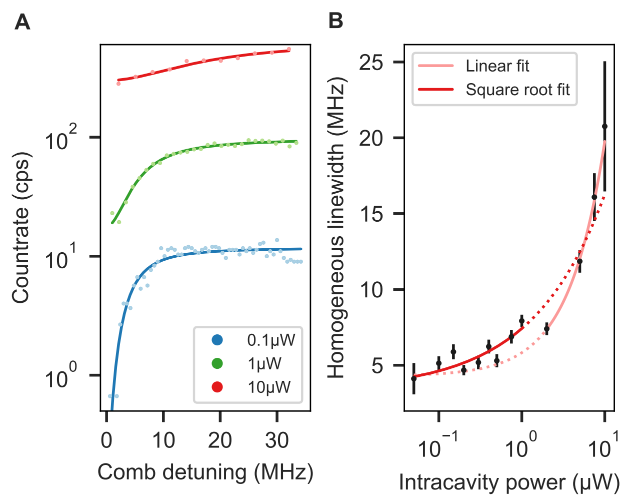

Fig. 6: A: transient spectral holes at three different intracavity power levels. The solid lines display fits of an inverted Lorentzian line. **B**: half-width of the transient spectral hole as a function of the intracavity power. Fitting the square root function of Eq. 2 (dark red) to the data permits to extrapolate an upper bound of the homogeneous linewidth. The four datapoints at highest power were excluded from the fit since a linear function (light red) yields better agreement.

since the homogeneous linewidth shows a linear temperature dependence below 10 K [20].

The measured linewidth is about a factor of two higher than the value of 1.6 MHz measured at 4 K by [23] using spectral hole burning spectroscopy in a large ensemble, and much larger than the value obtained from photon echo measurements (116 kHz) as shown in Fig. 6 in the supplementary information. The comparably broad linewidth can be explained by the high power levels above saturation that are required for this method. In this regime, instantaneous spectral diffusion (ISD) [30] due to dipole-dipole interactions between resonantly but also off-resonantly excited ions broadens the linewidth. Since all data points remain in this regime, an extrapolation to zero power can not yield a linewidth that is unaffected by ISD as one would expect it for frequency selective single ion addressing.

Using the measured linewidth, $\Gamma_h = 2\pi \cdot 3.3\,\mathrm{MHz}$, we are able to calculate the saturation intensity I_{sat} of a perfectly coupling single ion. Starting from the resonant saturation parameter given in [31] and expressing the Rabi frequency in terms of cavity QED parameters for a Fabry-Pérot type cavity mode, we arrive at the following expression:

$$I_{\text{sat}} = \frac{4\pi^3}{3}\hbar c \frac{\Gamma_h}{\zeta \lambda^3}. \tag{3}$$

This results in a value of $I_{\text{sat}} = 2 \cdot 10^4\,\mathrm{W/m^2}$ or a intracavity saturation power of $P_{\text{sat}} = 61\,\mathrm{nW}$ for the 580 nm transition at 4.3 K.

We note that with the performed spectroscopy, we already reached the level of the single ion saturation power and probe only a very small number of ions: We simulate the random distribution of the total number of ions (derived from the NP diameter and doping concentration) over the measured inhomogeneous profile taking into account the hyperfine structure to estimate the number of ions that we address within a power-broadened linewidth at the lowest power level of 50 nW intracavity power in Fig. 5 and 6. This yields about 15(2) europium ions.

6 Single ion count rate estimation

The agreement between the measured and estimated ensemble averaged Purcell factors in Fig. 4B and C as well as the confirmation of a homogeneous linewidth much narrower than the cavity linewidth (bad cavity regime) enables us to estimate the count rate that we expect from a perfectly coupling single europium ion. We simulate the count rate for pulsed, resonant excitation on the 580 nm transition with a 1 µs excitation pulse and variable length of the detection time window dependent on the repetition rate $R_{\text{rep}} = 1/(t_{\text{ex}} + t_{\text{det}})$. A 3-tone excitation pulse matching the ground state hyperfine splitting is assumed, to avoid optical pumping. Since we expect incoherent excitation, we assume an excited state population of $p_{\text{ex}} = 0.5$ for each trial. Further, we take the same cavity parameters as for the measurements presented above, i.e. a bare finesse of 17,500 (9,500) for the 580 nm (611 nm) transition, and a fiber radius of curvature of 25 µm resulting in mode waists of 1.41 µm (1.44 µm) at the double-resonance condition at about 6 µm cavity length, and 1.18 µm for contact mode at 2.5 µm cavity length. The cavity length jitter strongly affects the Purcell factor quantified by eq. (1) in [19] and thus the fluorescence count rate. Here, we set the cavity length stability to the best measured values of 2.5 pm (0.8 pm) for an open cavity (contact mode) as presented in [19]. The detected count rate can then be calculated as,

$$R_{\text{det}} = R_{\text{pulsed}}(d_{\text{NP}}) \cdot \eta_{\text{out}}(d_{\text{NP}}) \cdot T_{\text{path}} \cdot \eta_{\text{det}},$$

with the cavity outcoupling efficiency $\eta_{out} \leq 0.55$, which is dependent on the nanoparticle scattering losses. The detector has an efficiency of $\eta_{\text{det}} \approx 0.65$